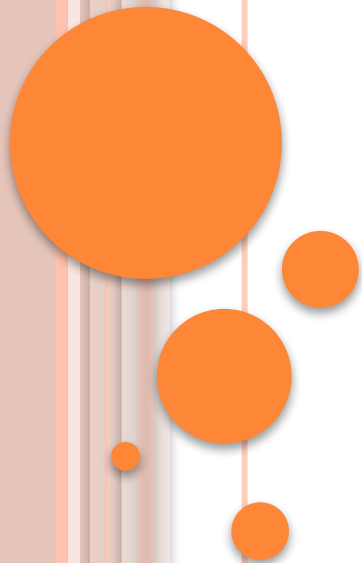


THROMBOSIS

Thrombus is blood that has clotted in the heart or a blood vessel



THREE CONDITIONS PREDISPOSE TO THROMBUS FORMATION

- **These are Virchow's triad**
 1. Endothelium injury
 2. Alterations in normal blood flow: *stasis* or *turbulence* of blood flow
 3. Hypercoagulable blood



ENDOTHELIAL INJURY

- Important for thrombus formation in the **heart or artery**
- The endothelium may or maynot be denuded
- Examples: Myocardial infarcts, Ruptured atherosclerotic plaques, inflammatory or traumatic vascular injury, Myocarditis sites, Inflamed or prosthetic cardiac valves, etc



ALTERATIONS IN NORMAL BLOOD FLOW

TURBULENCE AND STASIS

- Disturb the laminar blood flow
- Prevent dilution of activated clotting factors by fresh flowing blood
- Retard the inflow of clotting factor inhibitors
- Promote endothelial cell activation



- Turbulence contributes to **arterial & cardiac thrombosis by endothelial injury**
- Stasis is major factor for **venous thrombosis**
- Examples: Aneurysms, Atrial fibrillation, Myocardial infarcts, In dilated cardiac chambers, Over big ruptured atherosclerotic plaques



HYPERCOAGULABLE BLOOD

- Primary (genetic) or secondary (acquired) causes
- **Genetic**: mutations in the Factor V & prothrombin genes, etc
- **Acquired**: Prolonged bed-rest or immobilization, tissue damage (surgery, fracture, burn), cancer, pregnancy, oral contraceptive use, smoking, sickle cell disease, etc



MORPHOLOGY: THROMBUS

- Any where in cardiovascular system
- Variable in size & shape



MORPHOLOGY

- Ante-mortem thrombi tend to be rather **hard** and **crumbly or friable**, **area of attachment**
- Post mortem clots **do not have lines of Zahn**
- They are **rubbery**, **gelatinous**, and **non-adhesive**



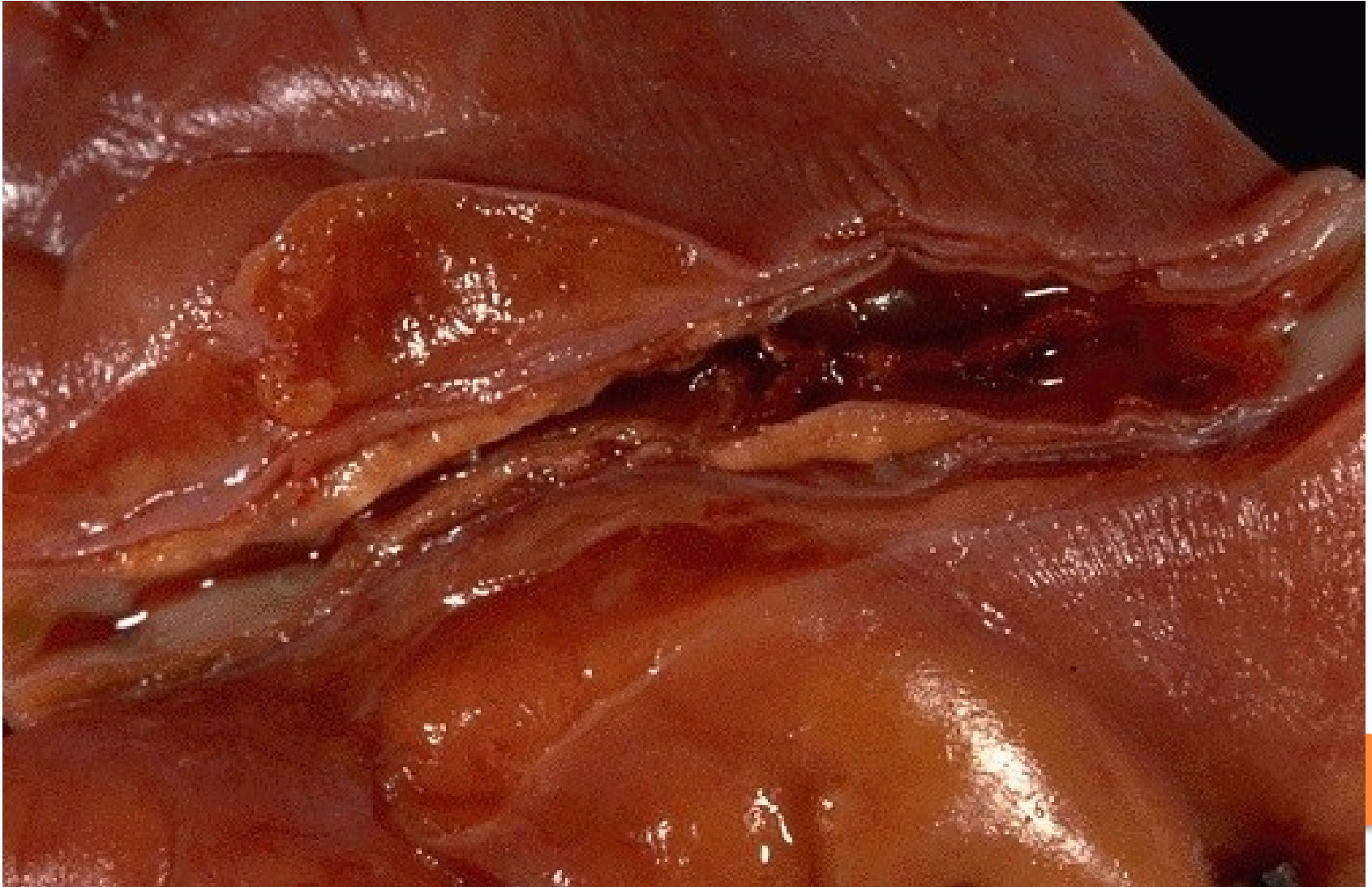
AORTIC ATHEROSCLEROSIS WITH THROMOBUS



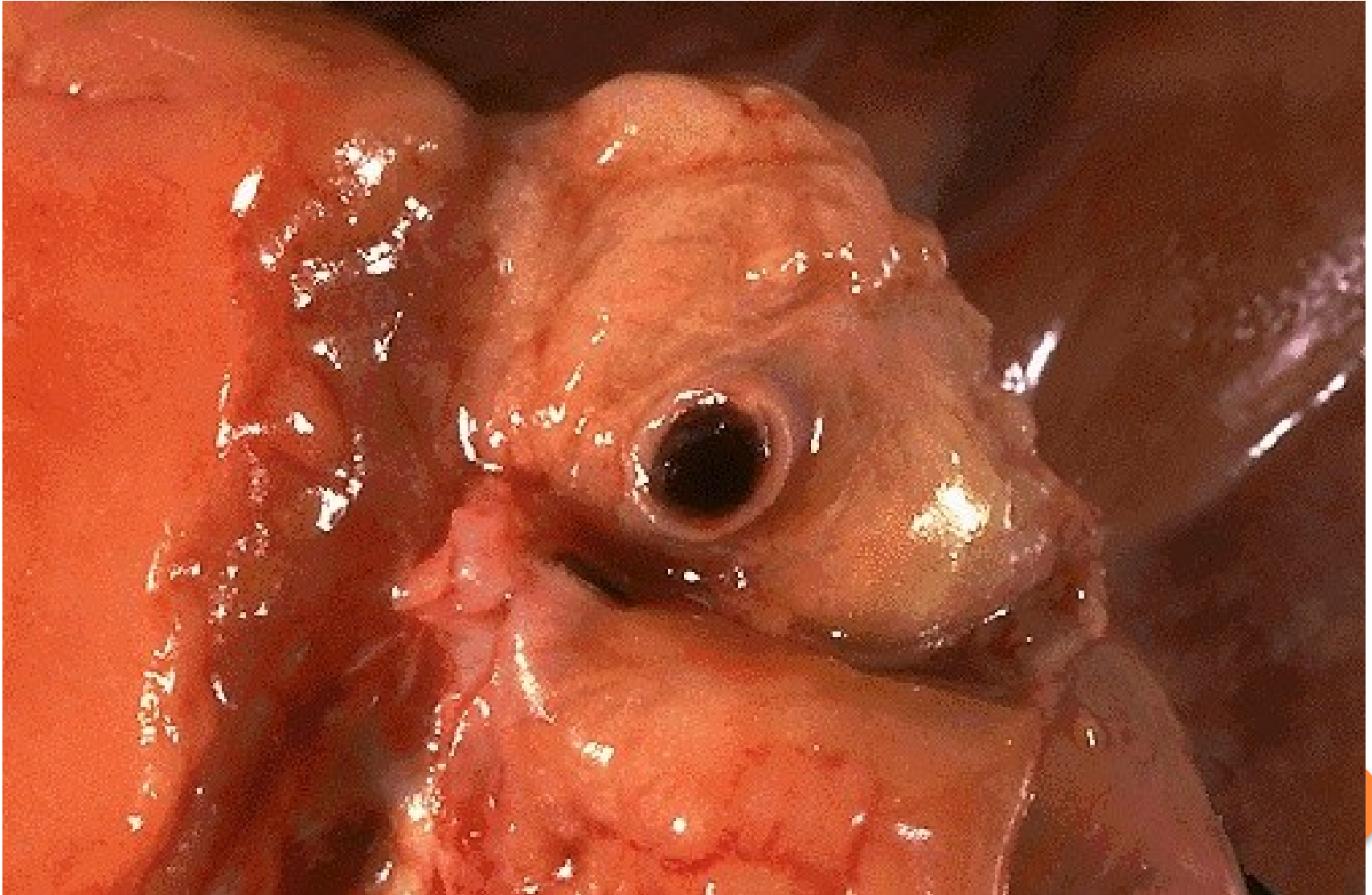
AORTIC ATHEROSCLEROSIS WITH THROMOBUS



THROMBOSIS OF A CORONARY ARTERY



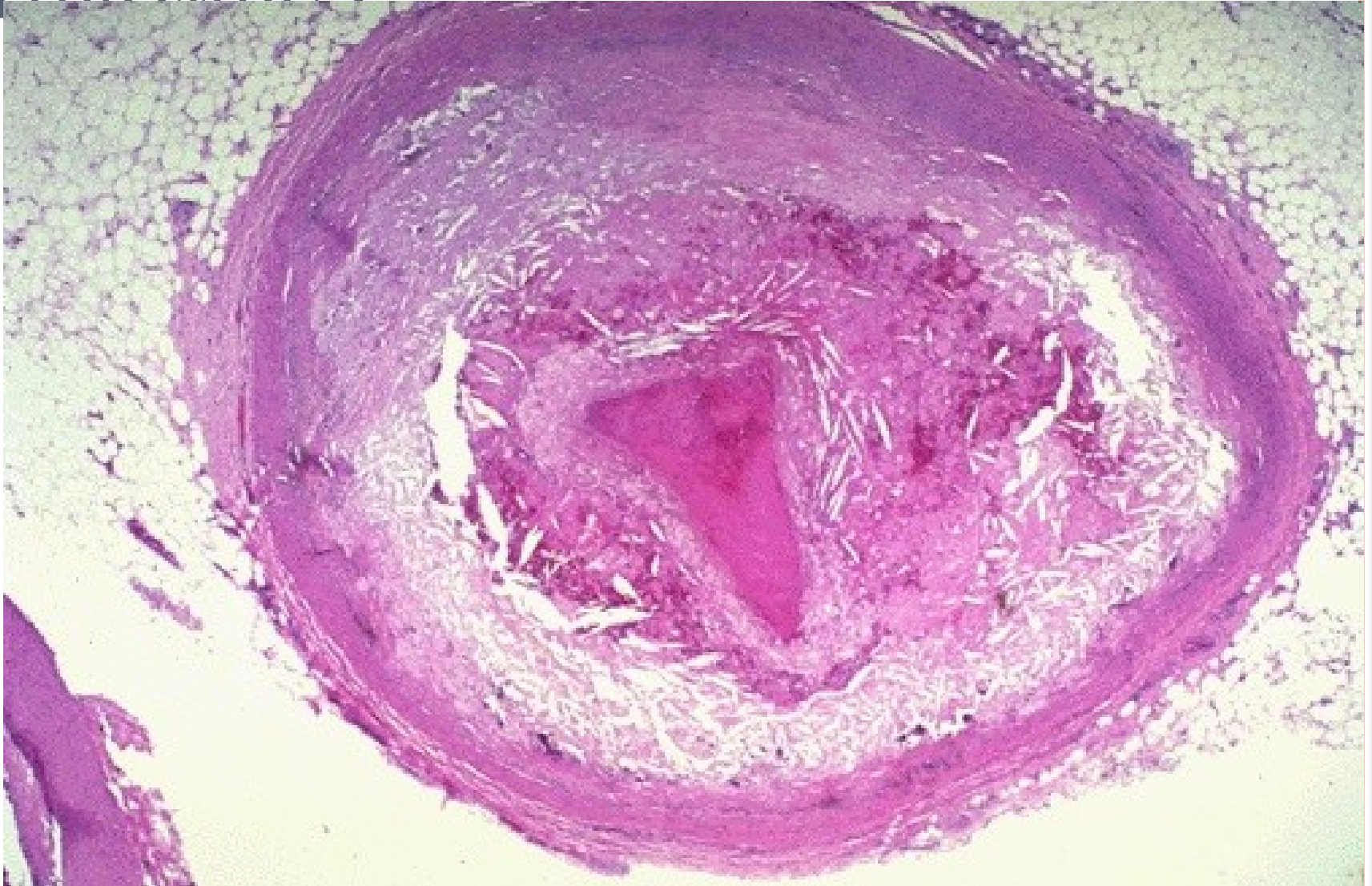
THROMBOSIS OF A CORONARY ARTERY



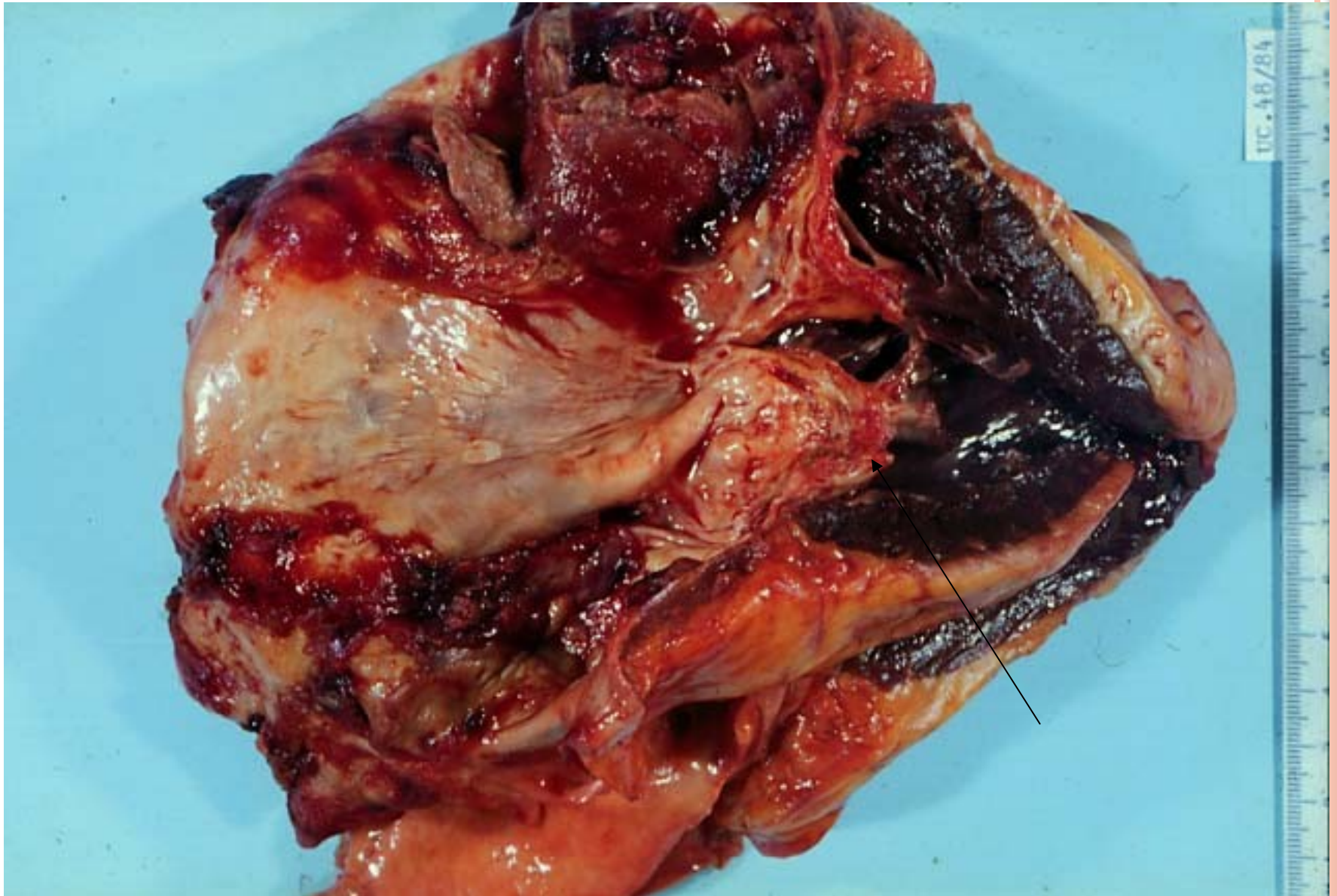
INFERIOR VENA CAVA THROMBOSIS



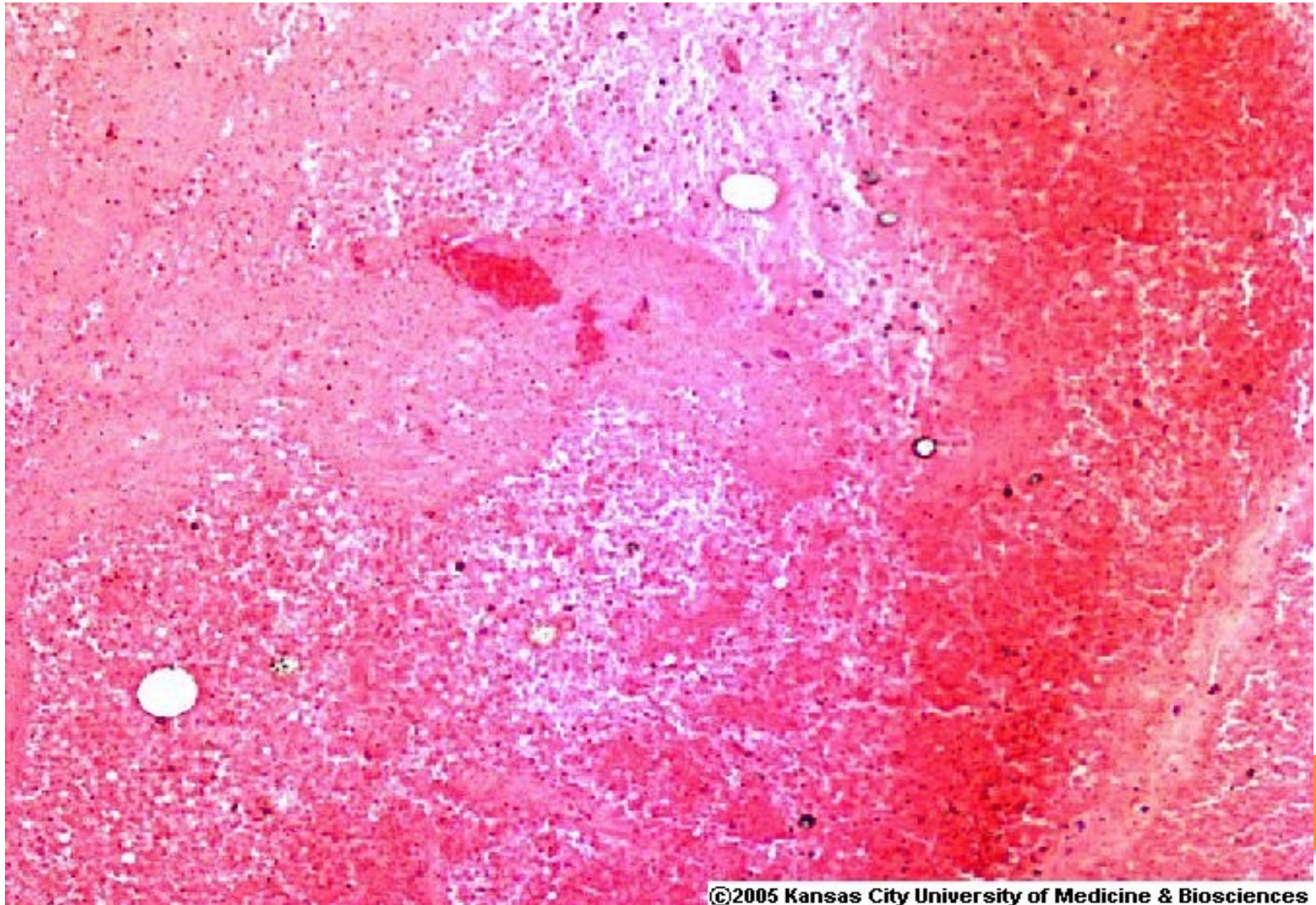
MIC: ATHEROSCLEROSIS WITH THROMBOSIS



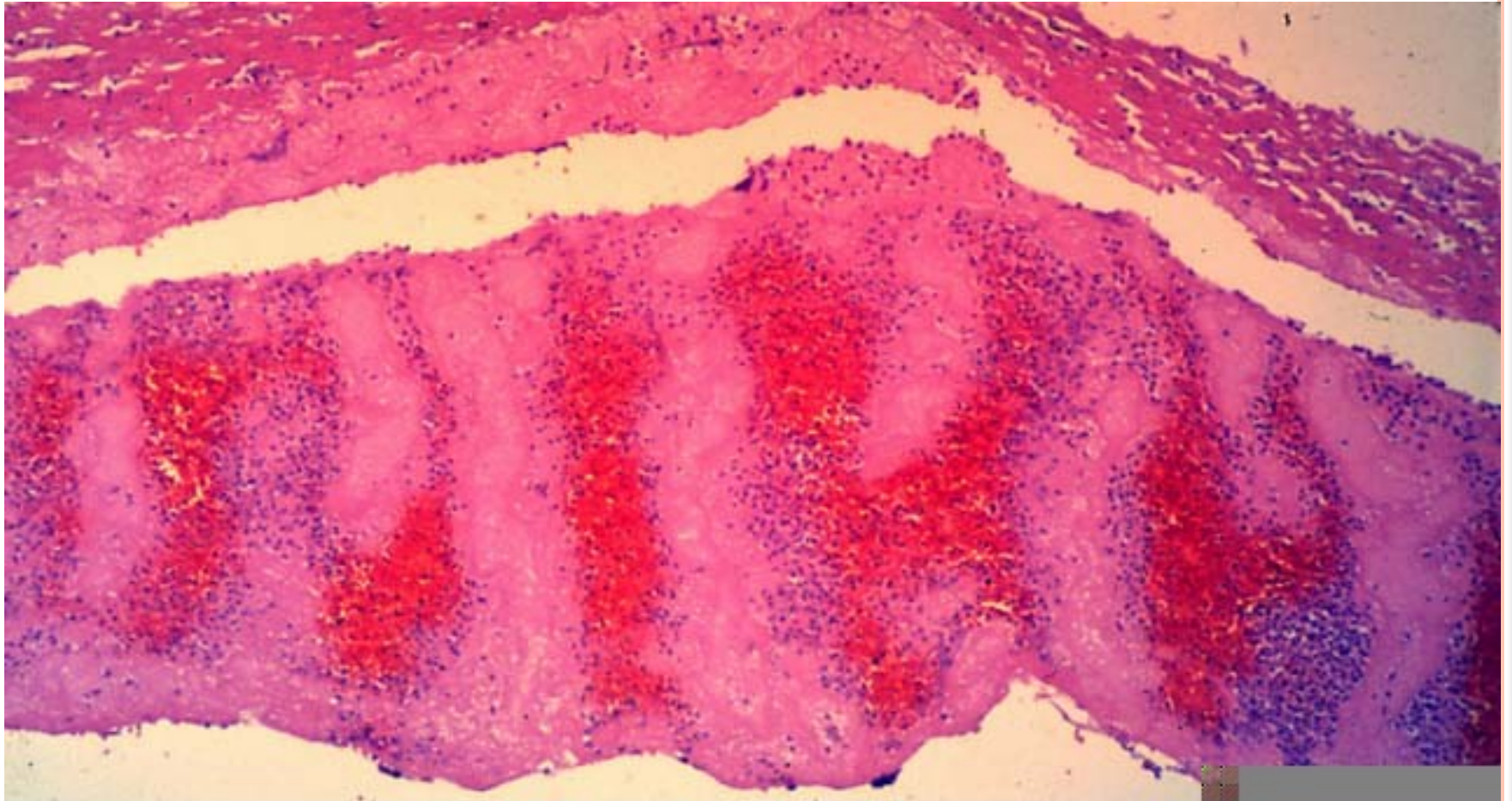
ATRIAL THROMBUS



THROMBUS



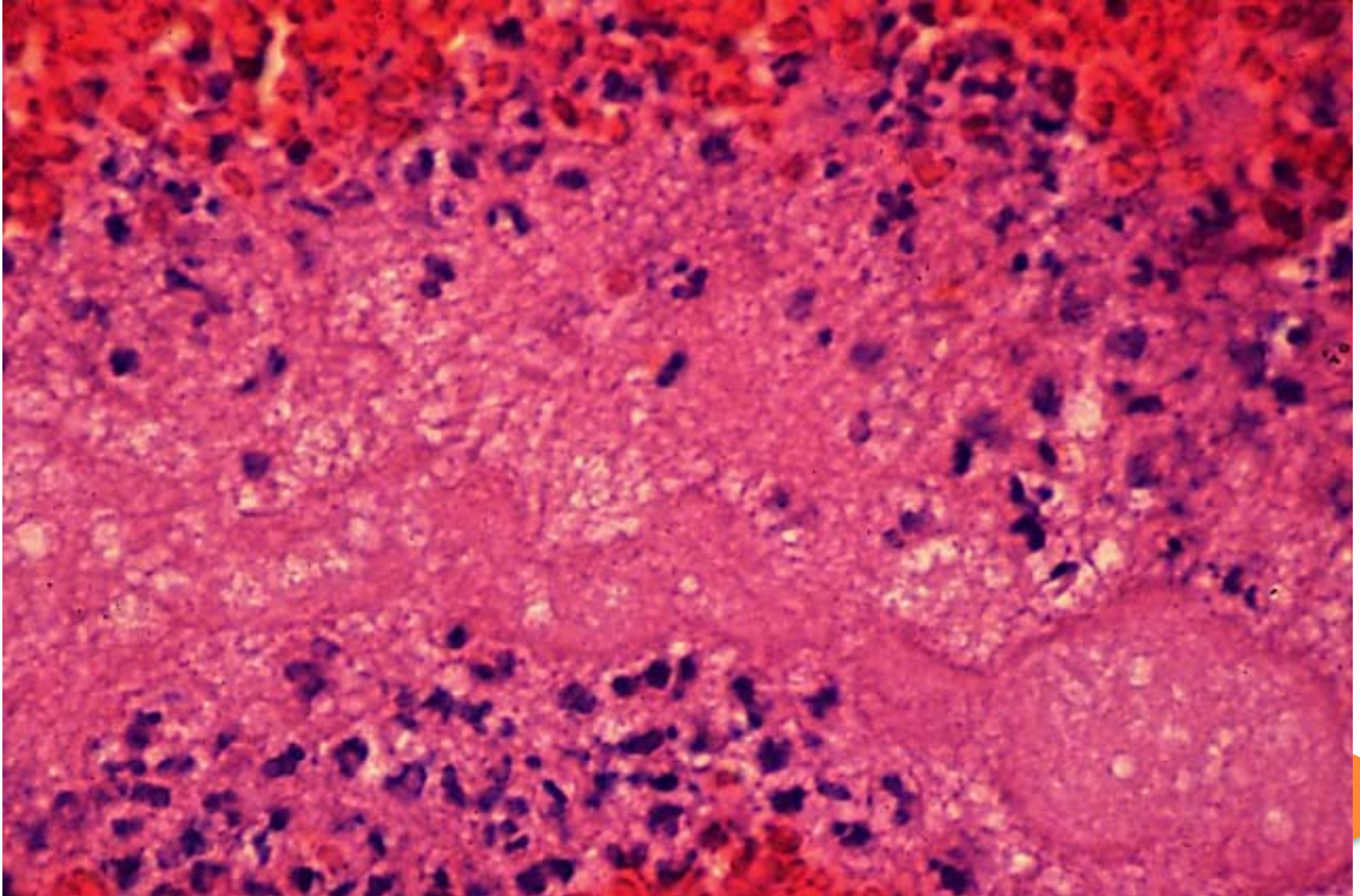
LINES OF ZAHN



LINES OF ZAHN



LINES OF ZAHN



POST MORTEM CLOT



SPECIAL TYPES OF THROMBI

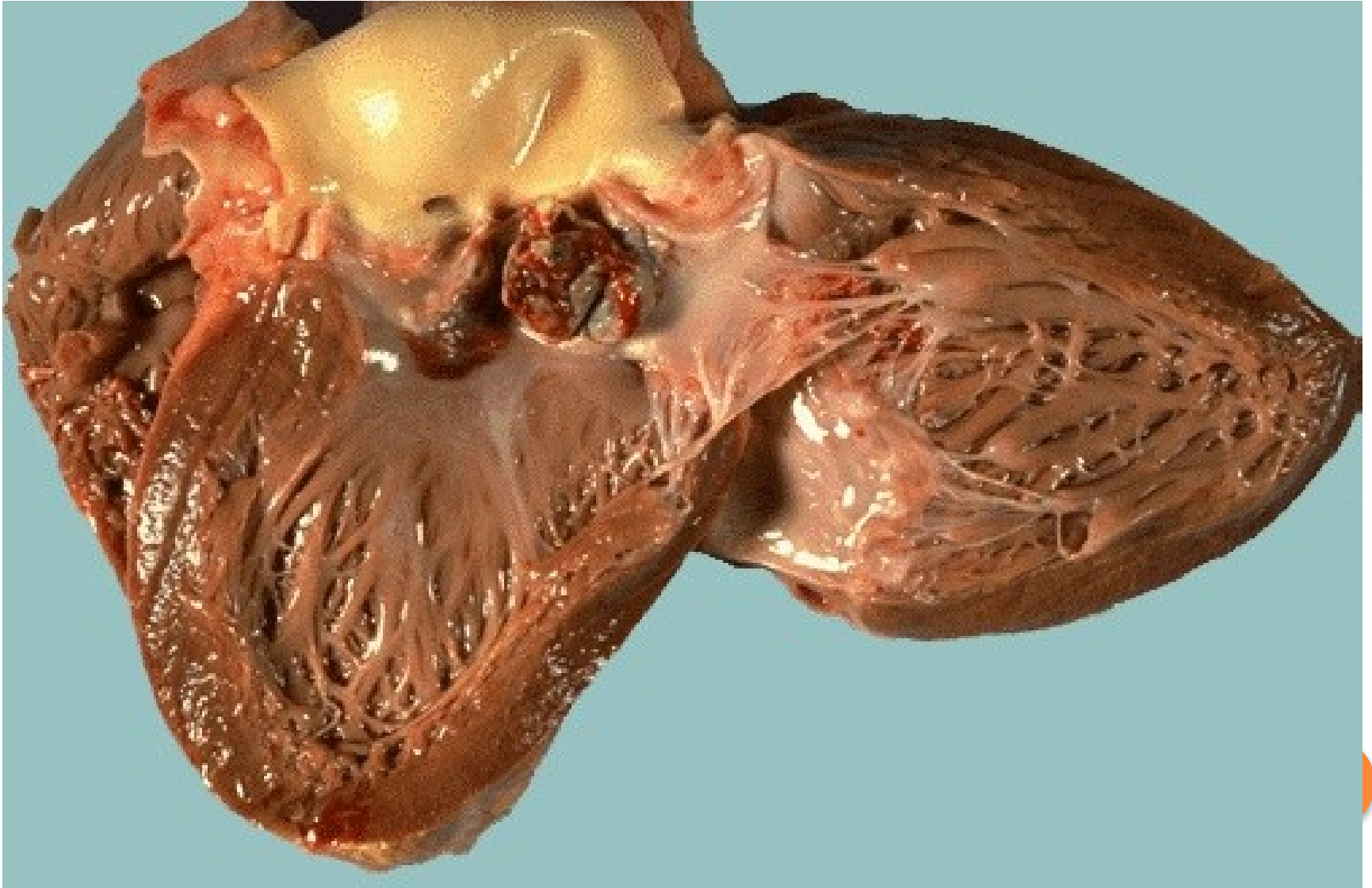
- **Mural thrombi:** on the walls of the cardiac chambers and aorta & do not occlude the lumen
- **Arterial thrombi** are usually **occlusive**
- **Vegetations** are thrombi that occur on cardiac valves. They may be loaded with bacteria (bacterial endocarditis), or sterile (bland).



- **Venous thrombi** (phlebothrombosis) almost always occlude the vein & are red
- 90 % in veins of lower extremities



AORTIC VALVE VEGETATION



FATE OF A THROMBUS

1. Propagation: as more clot forms on their surface. It is in the direction of blood flow, and may ultimately obstruct the blood flow
2. Embolization: fragments or breaks free and travel to other sites in the vasculature => producing **thromboemboli**
3. Dissolution (for recent thrombi): plasmin => **fibrinolysis**
4. **Organization** (for older thrombi)
5. **End recanalization**



OUTCOMES OF VENOUS THROMBOSIS

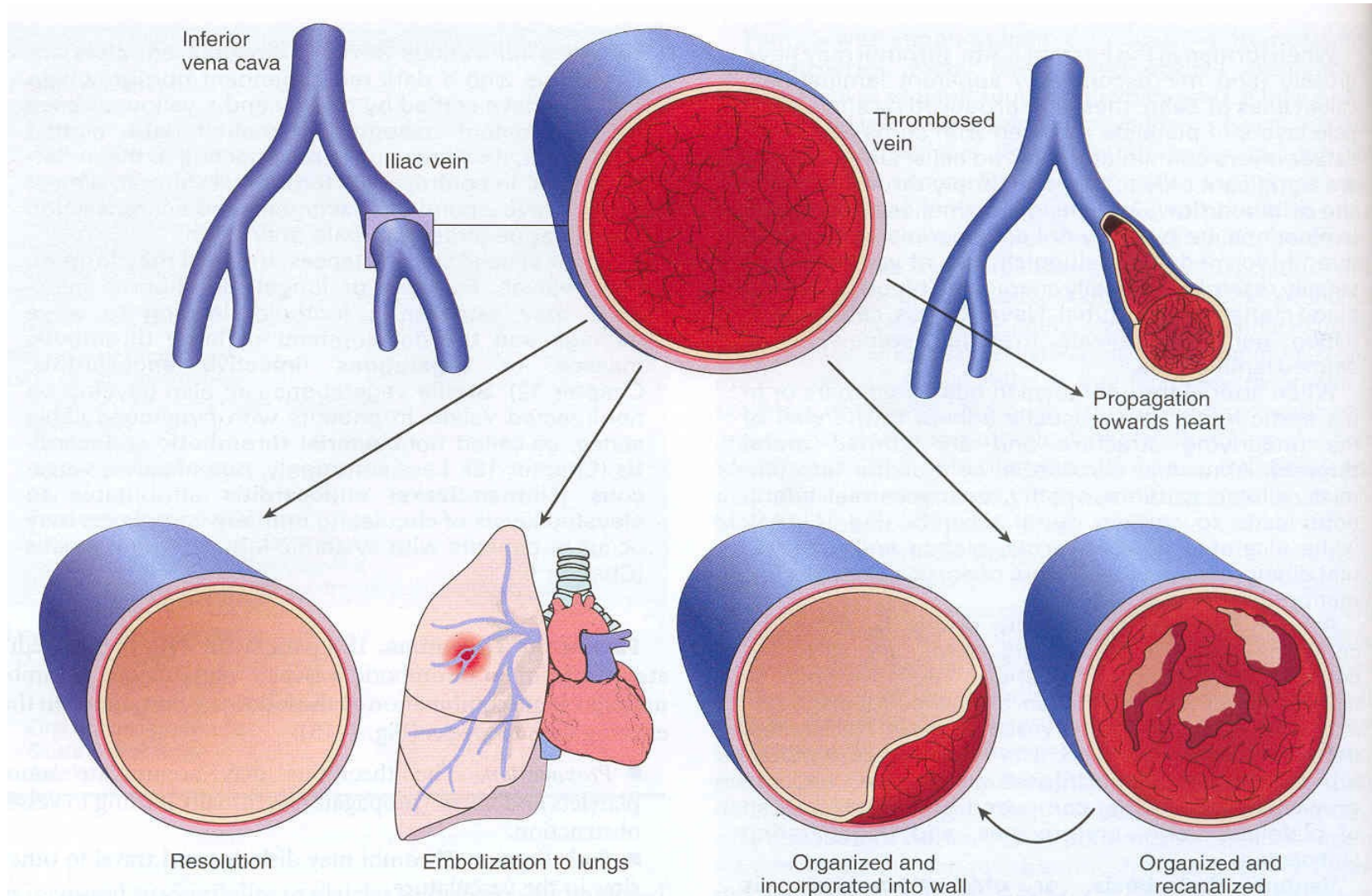
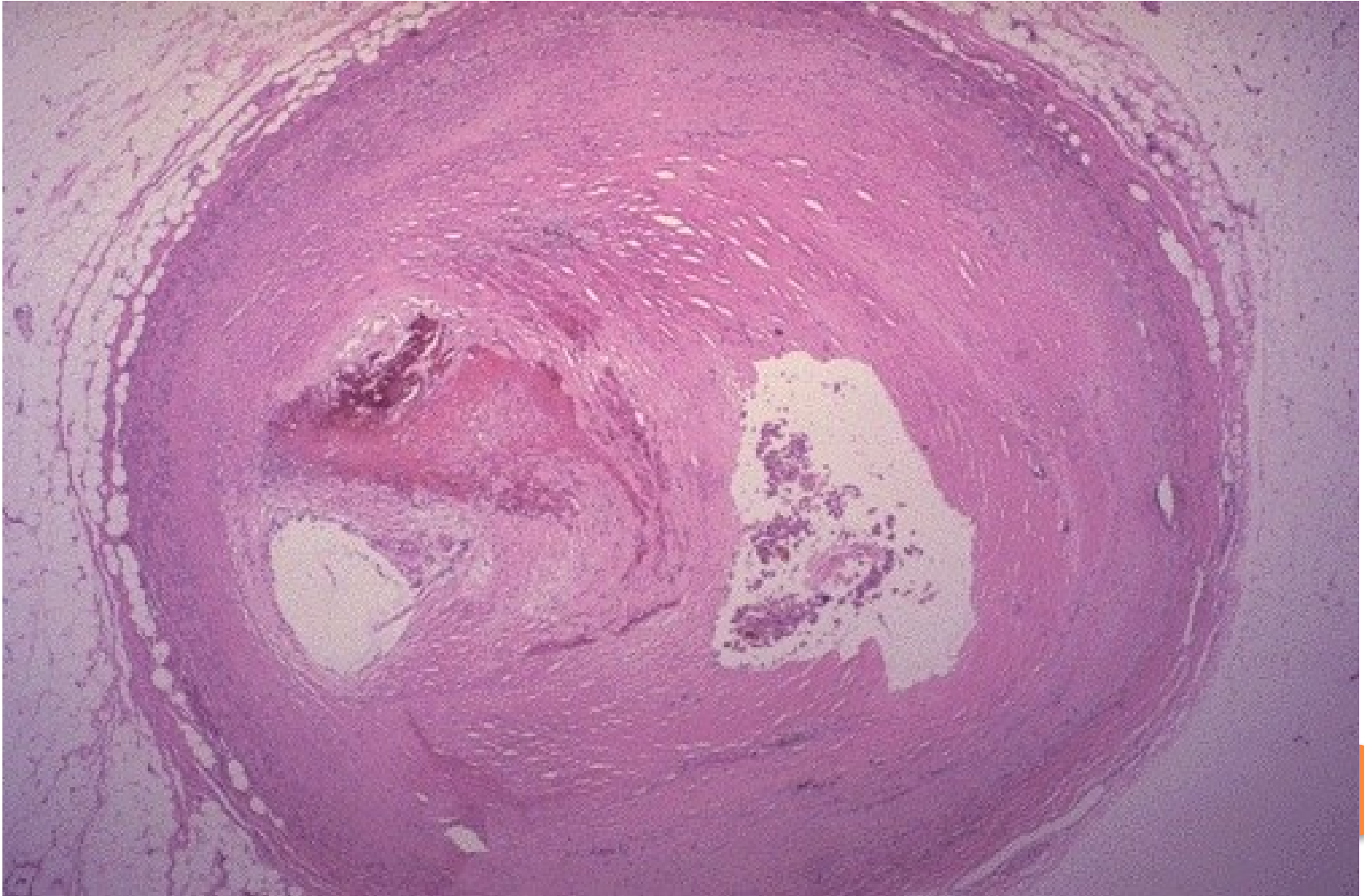


FIGURE 4-15 Potential outcomes of venous thrombosis.

RECANALIZED THROMBUS



CLINICAL IMPLICATION

- Thrombi can cause **obstruction of arteries & veins**
- Thrombi are possible source of **emboli**
- Deep Vein thrombosis (DVT)
- Arterial & cardiac thrombosis



EMBOLISM

- **Embolus**: any solid, liquid, or gaseous material that travels along the bloodstream.
- It will **impact and lodge** somewhere => partial or complete vascular occlusion => **infarction**
- "Embolus" comes from the Greek for "bottle stopper"



- **Embolism** is the clinical syndrome of having one or more emboli
- Most emboli are dislodged thrombi i.e., **thromboemboli**
- Pulmonary or systemic thromboembolism



PULMONARY THROMBOEMBOLISM

- > 95% of pulmonary emboli come from the **deep leg veins**
- A **paradoxical embolus** is one from the systemic veins that passes through a right-to-left intracardiac shunt to occlude a systemic artery.
- The **size** & the **cardiopulmonary status** of the patient determine the outcome



OUTCOMES OF PULMONARY THROMBOEMBOLISM

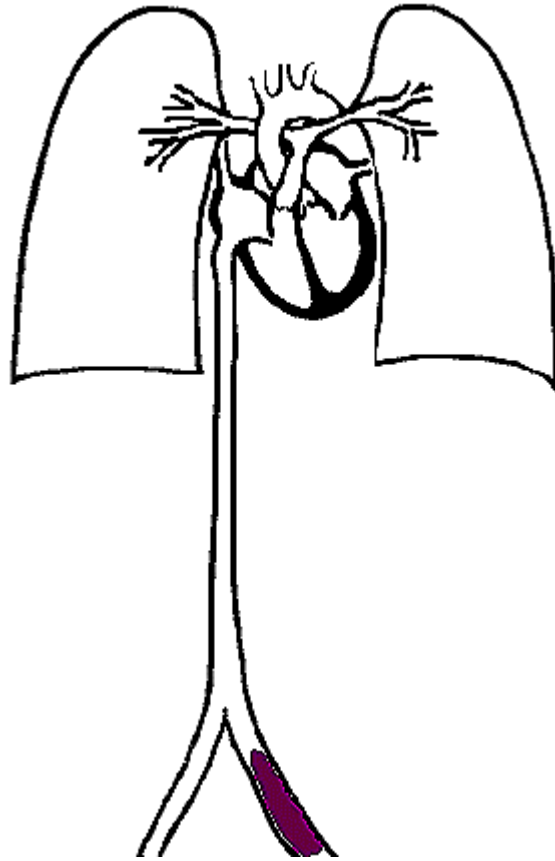
1. 60-80% are **asymptomatic** because they are small => organization
2. Sudden death, right heart failure or cardiovascular collapse occurs when **60% or more of the pulmonary circulation is obstructed**
3. **Pulmonary hemorrhage**: obstruction of medium sized arteries



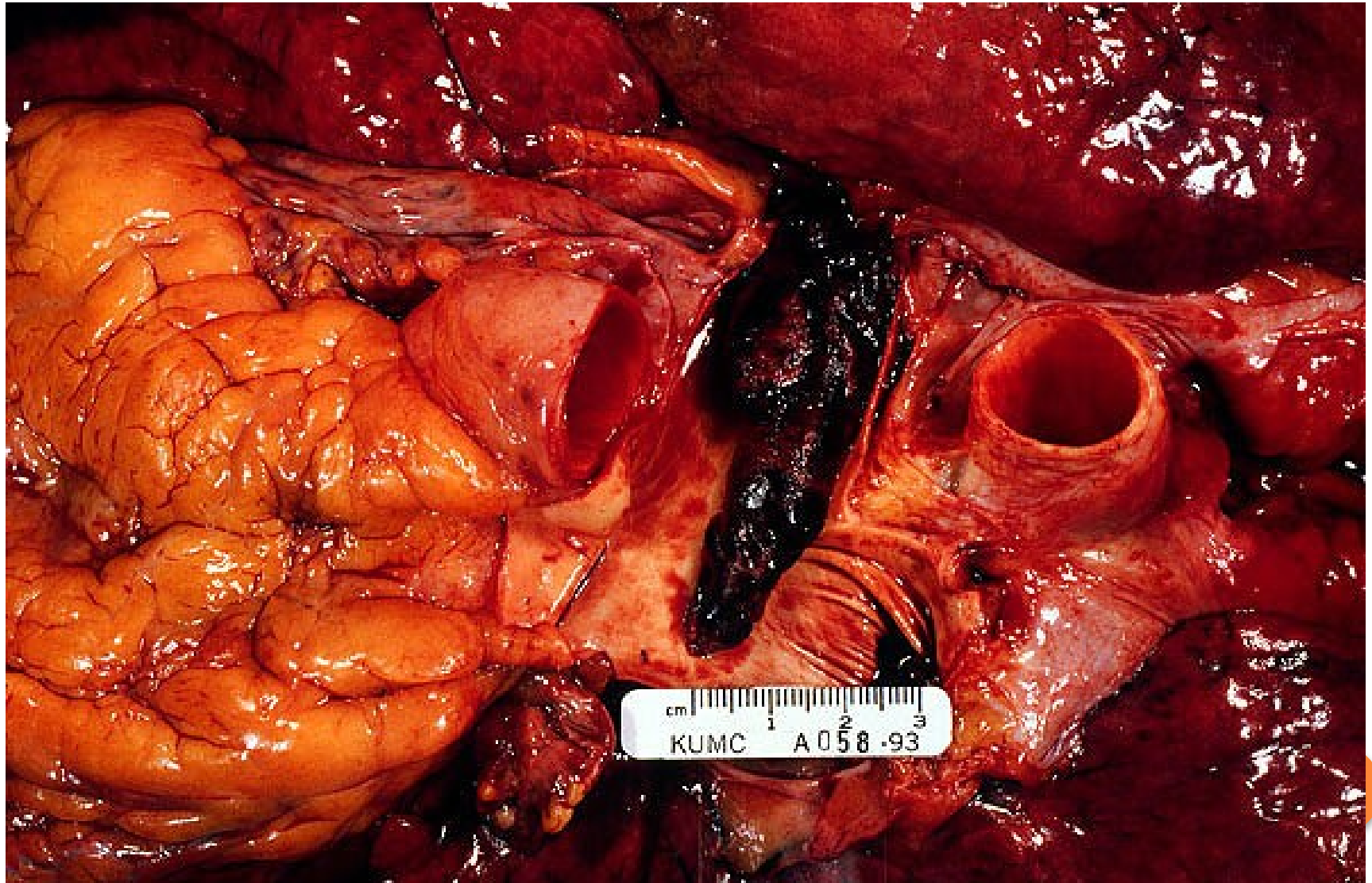
OUTCOMES OF PULMONARY THROMBOEMBOLISM CTD

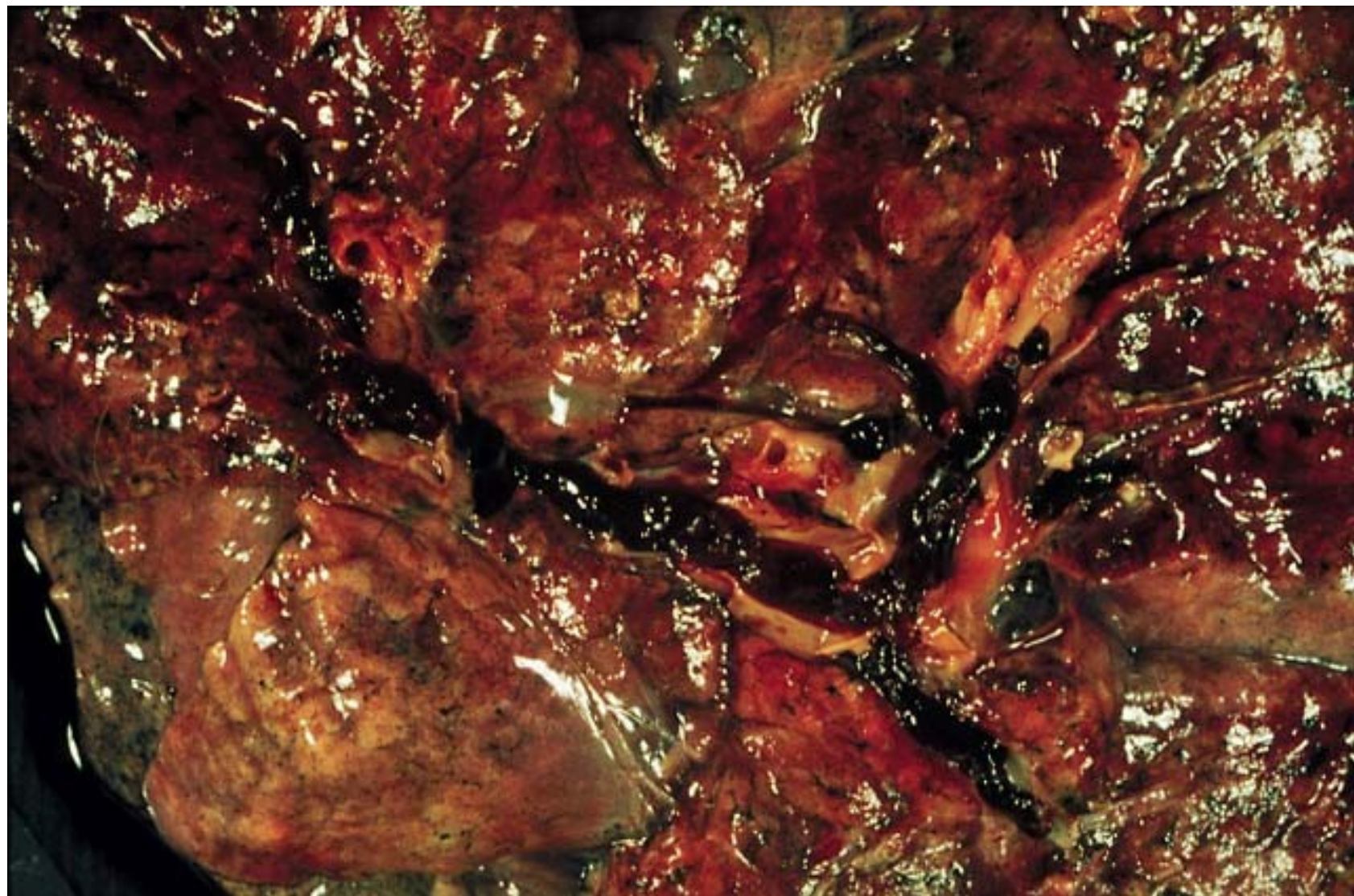
- 4. Pulmonary infarction in a patient with left sided heart failure
- 4. Multiple emboli over time may cause pulmonary hypertension



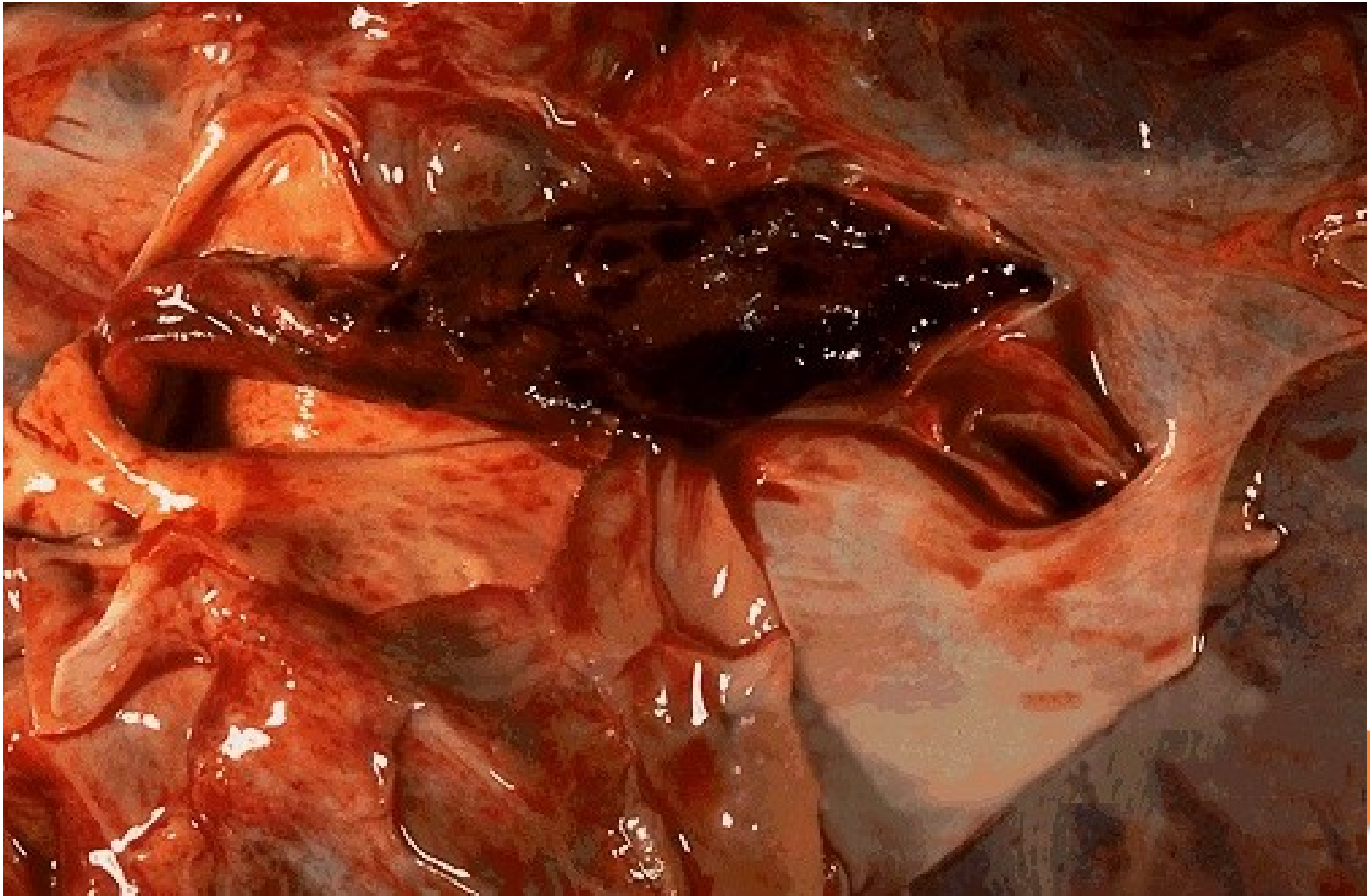


PTE





SADDLE PULMONARY THROMBOEMBOLISM



SYSTEMIC THROMBOEMBOLISM

- Emboli traveling within the **arterial circulation**
- **80%** arise from **intracardiac mural thrombi** & 10-15% origin unknown
- No artery is immune, though the majority (**75%**) go to the **lower extremities**, **10% brain**
- In contrast to pulmonary emboli, systemic emboli **almost always produce infarcts**.



- Note that the **emboli of bacterial endocarditis** are laden with micro-organisms, and that **abscess formation** is likely
- The consequence depends on the **extent of collateral vascular supply**, the **tissue vulnerability to ischemia**, & the **caliber of the vessel occluded**





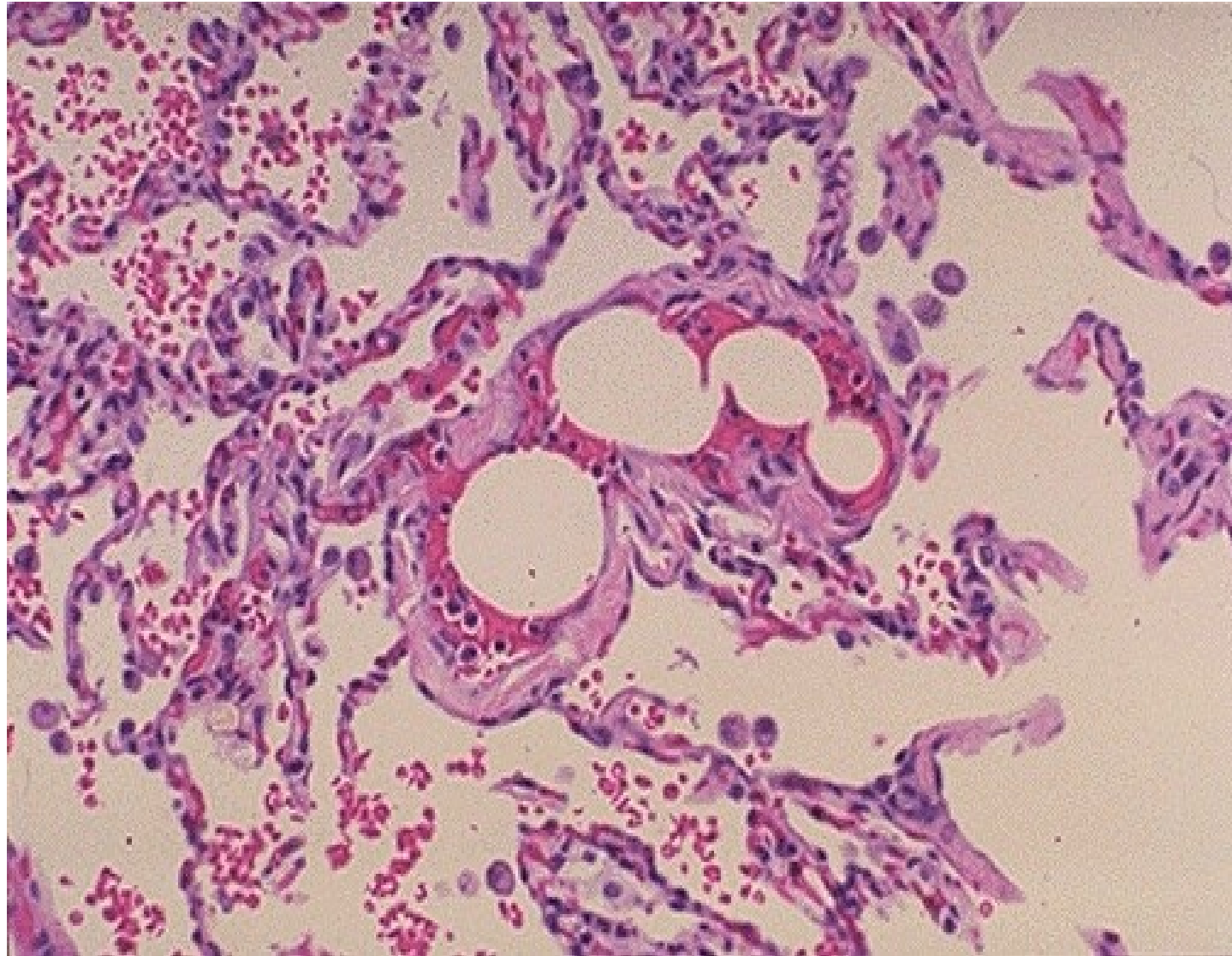
TYPES OF EMBOLISM

FAT EMBOLISM

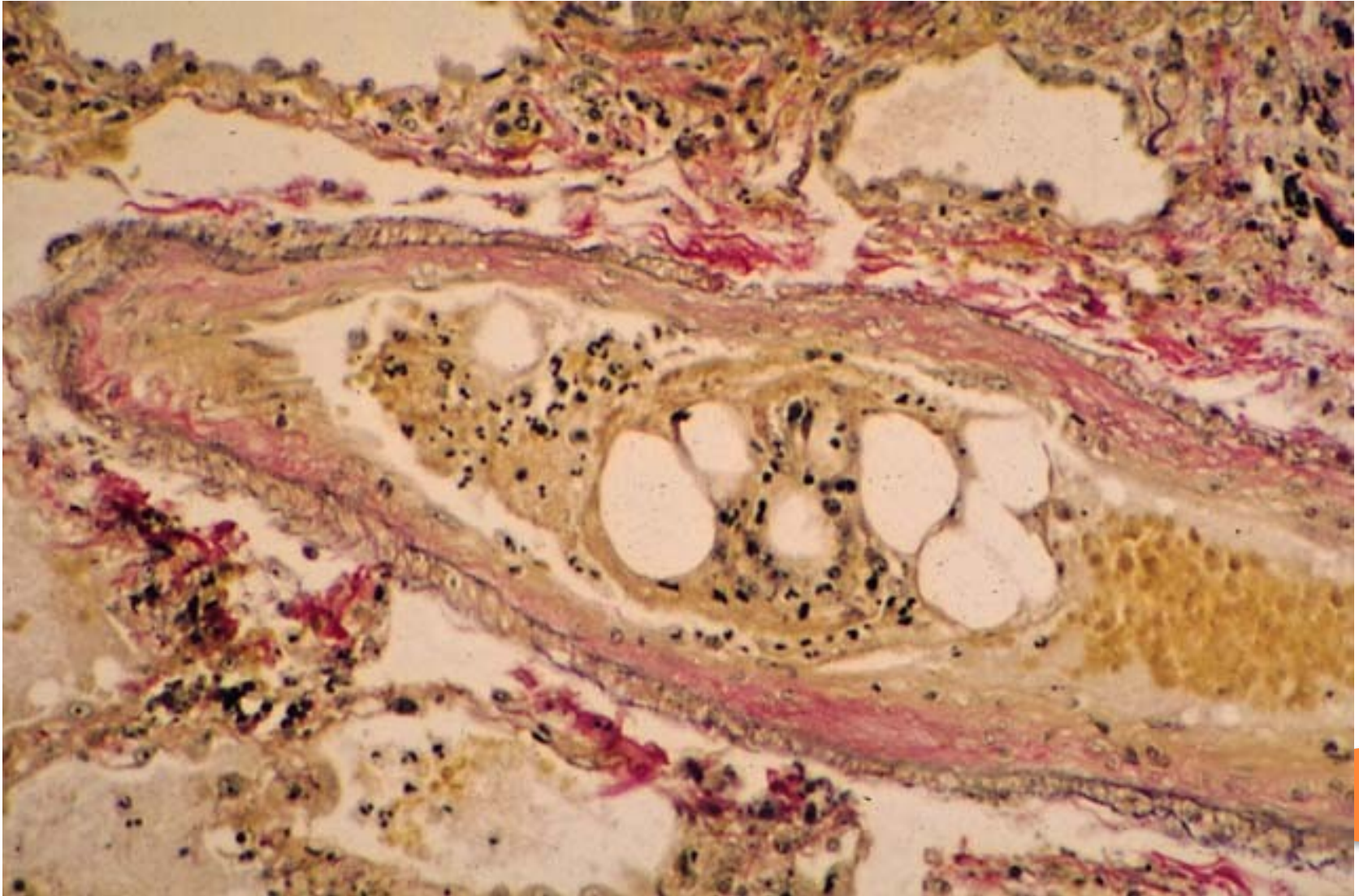
- After fractures of bones, **globules of fat** are released into the circulation.
- Occurs 1-3 days after injury with heavy fat embolization
- **<10% are symptomatic**
- **Mechanical & biochemical injury**



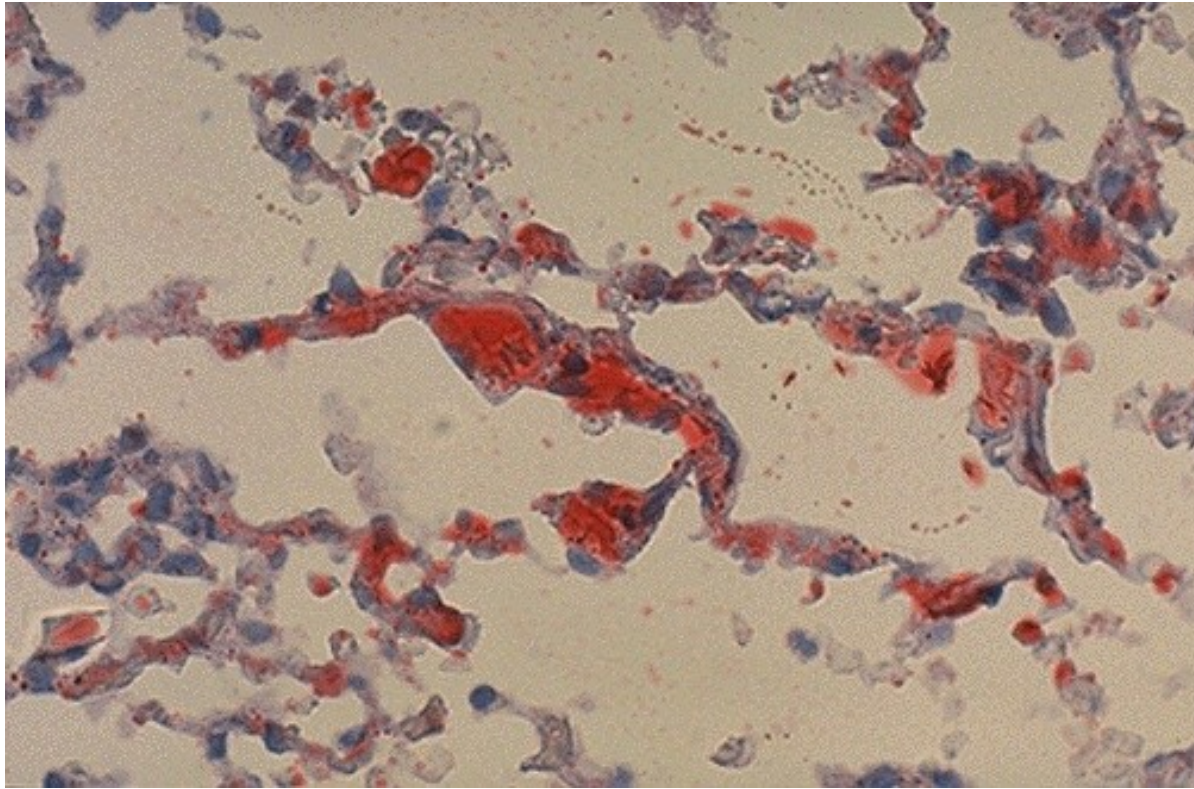
FAT EMBOLISM



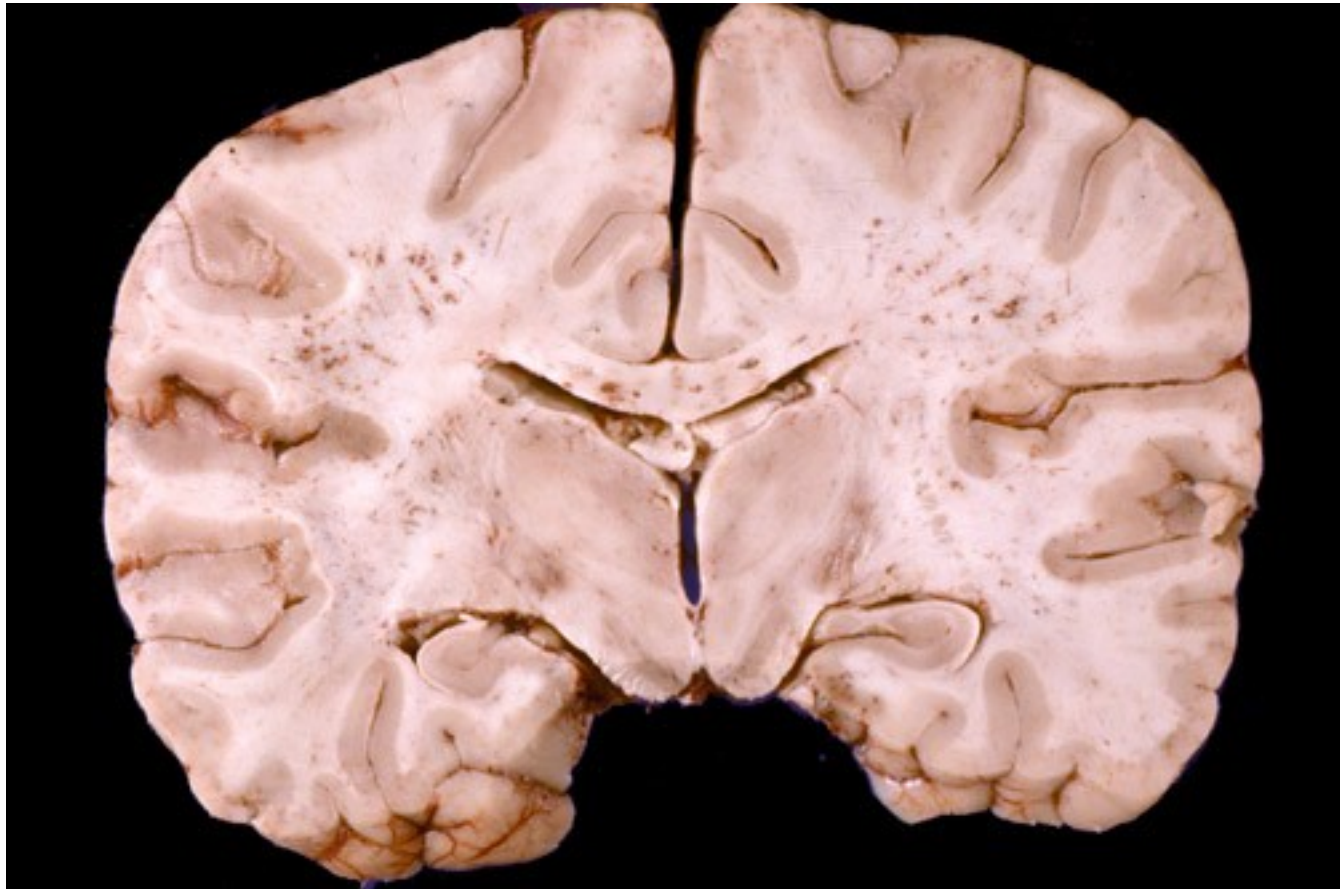
FAT EMBOLISM



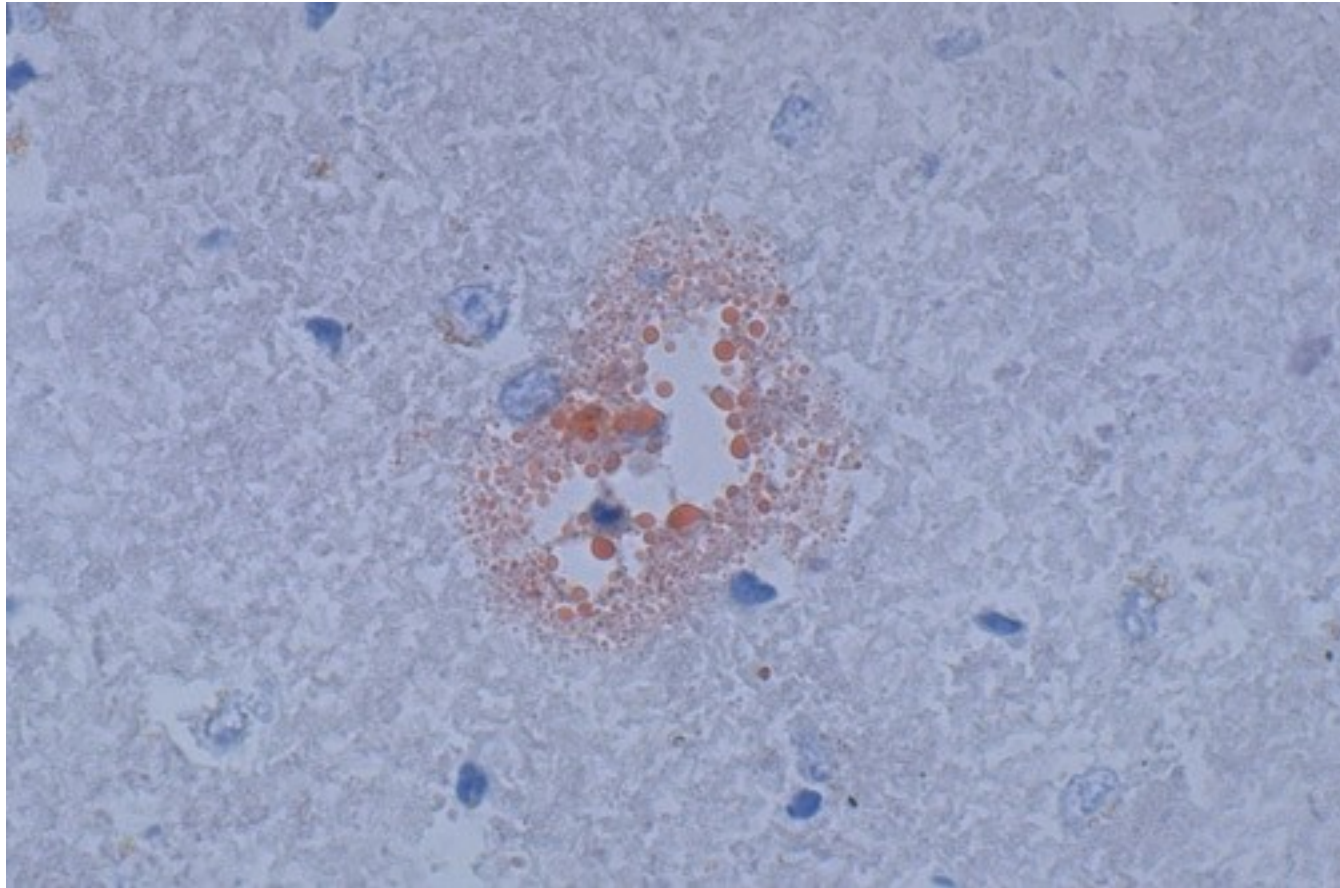
OIL RED SPECIAL STAIN



FAT EMBOLISM: PETECHIAL HEMORRHAGE



FAT EMBOLISM: BRAIN

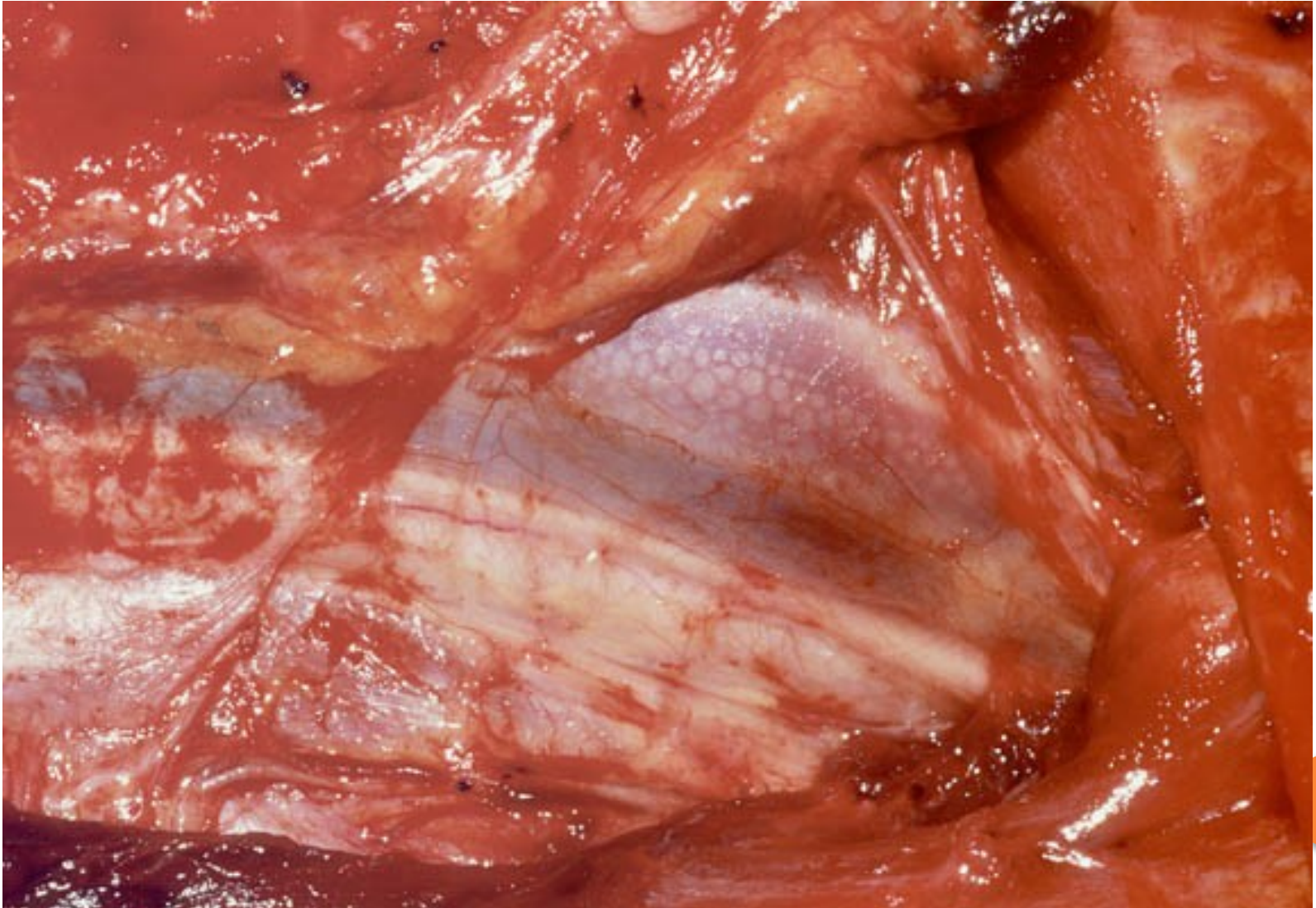


AIR EMBOLISM

- Gas in the circulation
- > 100cc produce clinical effect by forming bubbles
- Causes: ruptured uterine veins & chest trauma, etc



AIR IN JUGULAR VEINS AUTOPSY

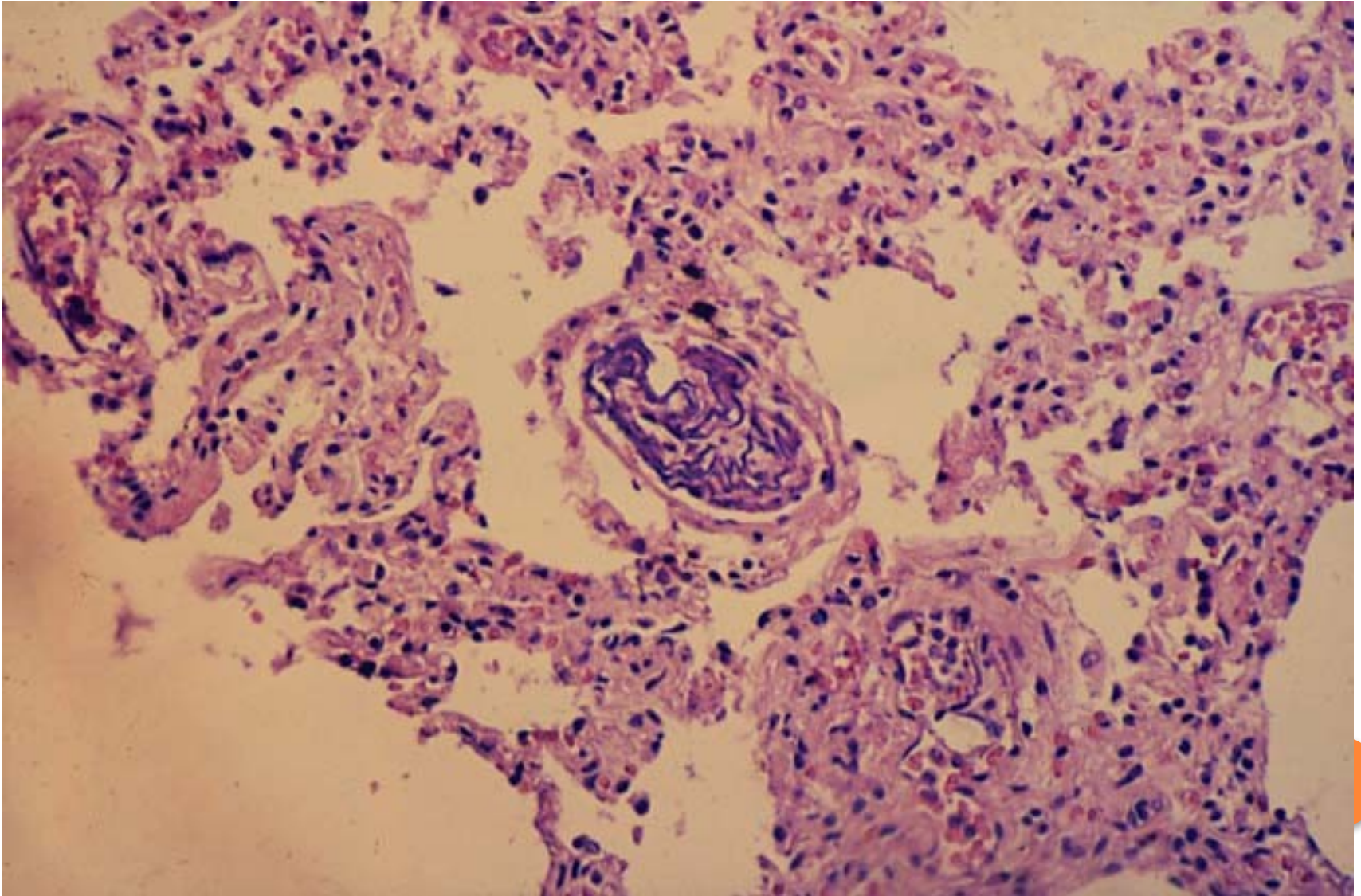


AMNIOTIC FLUID EMBOLISM

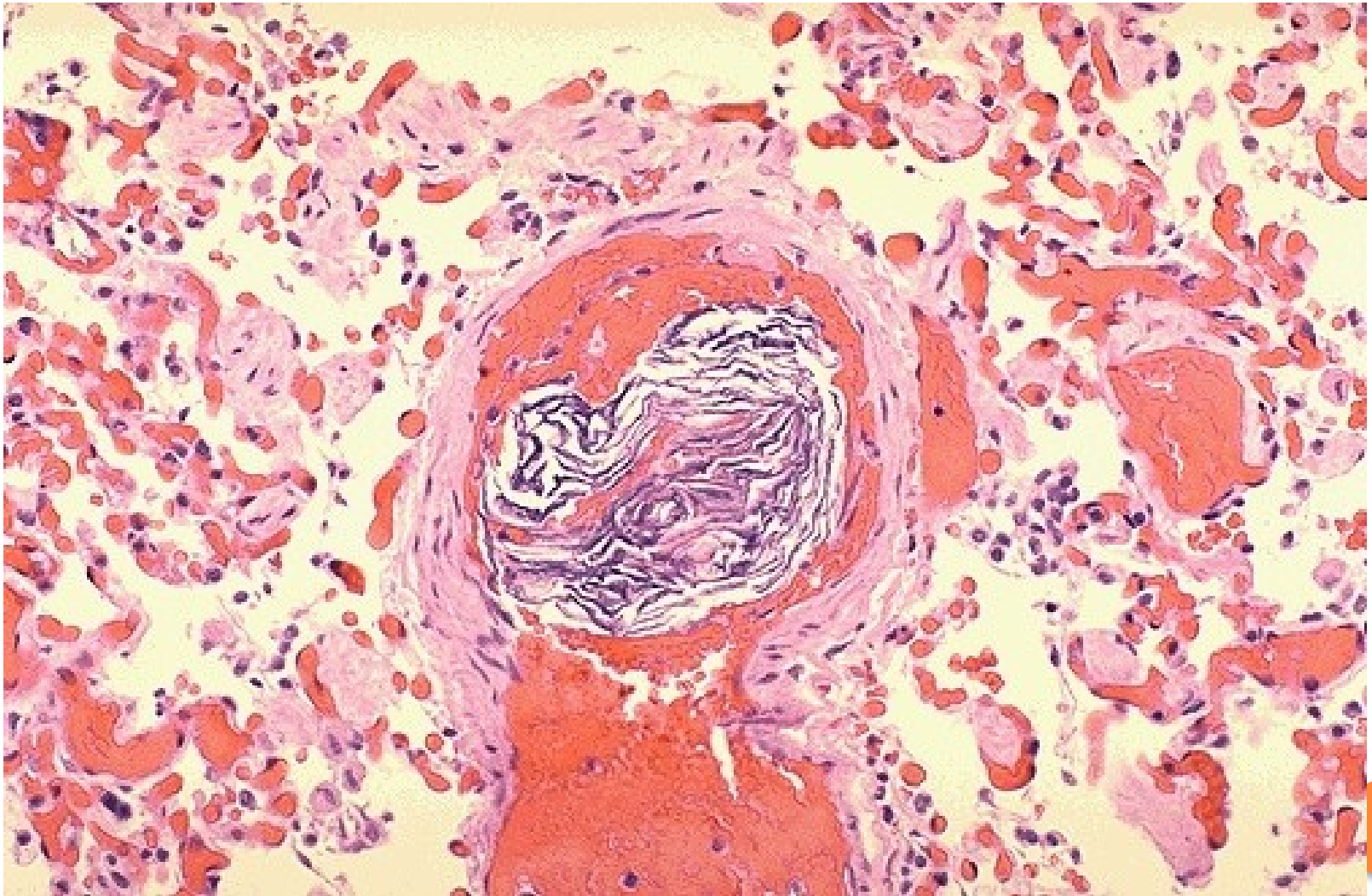
- It occurs in 1 out of 50,000 deliveries
- **Amniotic fluid**, which is full of baby's debris, enters the systemic circulation
- **Respiratory difficulty** and **shock** are followed by **DIC**
- **Squamous cells** and other debris in the pulmonary vasculature.
- **20-50%** mortality rate



AMNIOTIC FLUID EMBOLISM



AMNIOTIC FLUID EMBOLISM

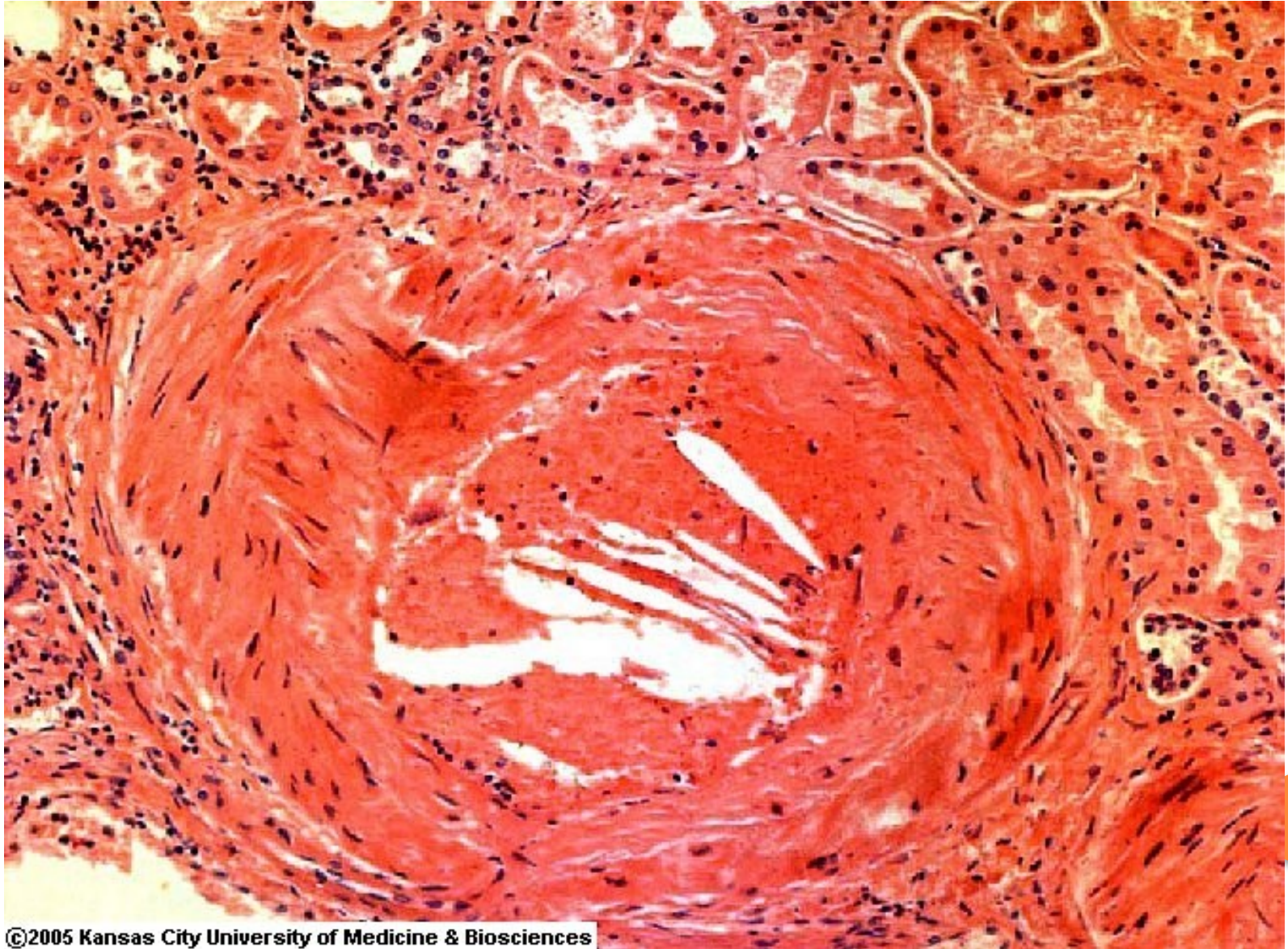


OTHER TYPES OF EMBOLISM

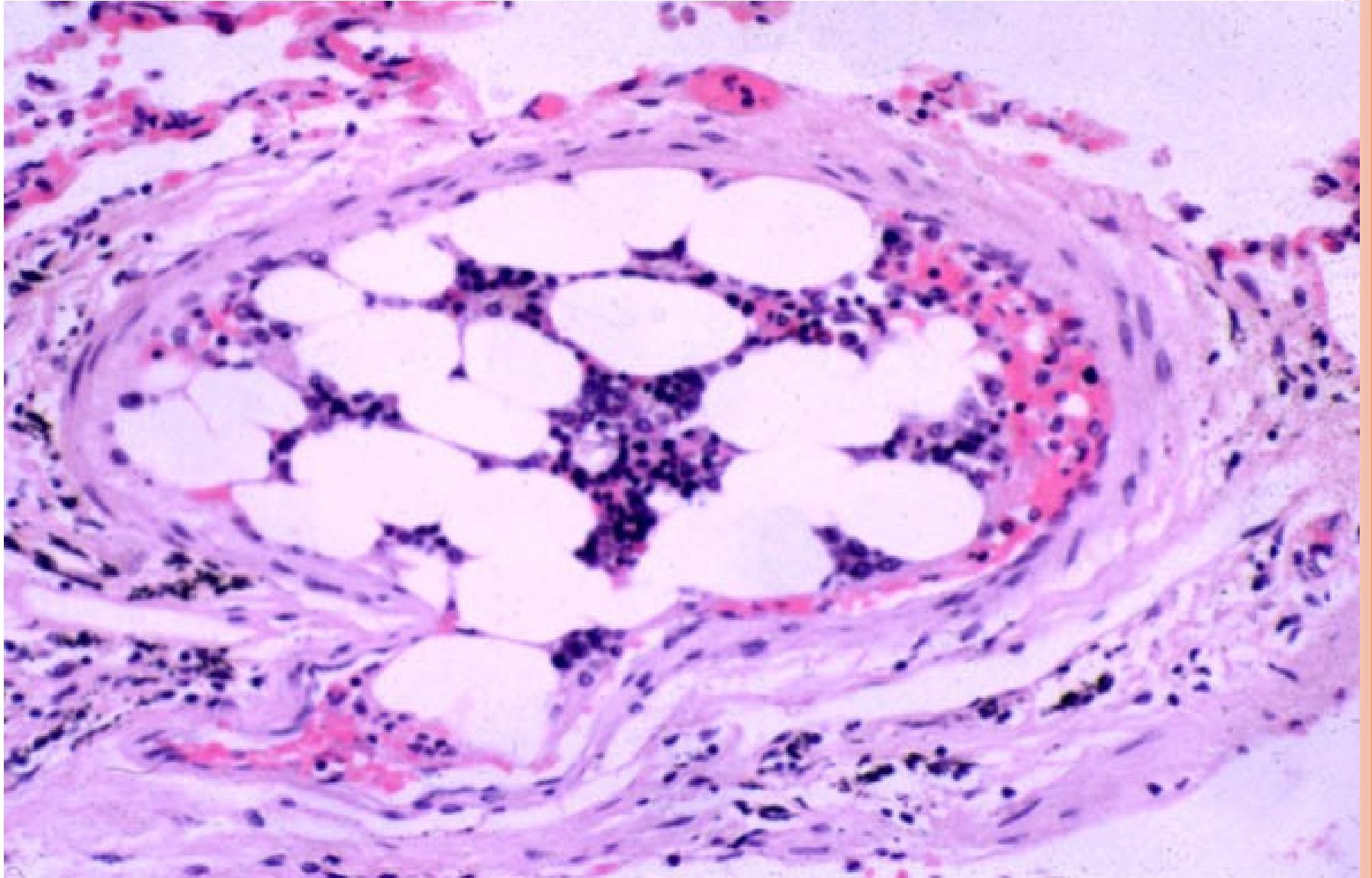
- **Atheroemboli** are bits of debris from inside atherosclerotic plaques.
- **Bone marrow emboli** with fractures of ribs
- **Tumor emboli** are bits of cancer that invaded a vein and then broke off eg. Renal cell carcinoma



ATHEROEMBOLI



BONE MARROW EMBOLI



- Reading Assignment

Mechanism of Hemostasis---primary and Secondary.

